

LABSHEET - 2**HIMANI CHAUDHARY****BTECH(CSE) 3RD YEAR****CU240250967****SECTION - B****ROLL NO. - 24****Mastery in Data Analytics and Visualizations and Career Advancement (Technical)**

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

```
# 1. LOAD IMAGE
# =====

image_path = "/content/beautiful-nature-mountain-scenery-with-flowers-picjumbo-com.jpg"
# Actually load the image
image = cv2.imread(image_path)

if image is None:
    print("ERROR: Image not found!")
    print("Check the image path and filename.")
    exit()
else:
    print("Image loaded successfully!")
    print("Himani Chaudhary")
    print("CU240250967")
```

Image loaded successfully!
Himani Chaudhary
CU240250967

```
# 2. CONVERT IMAGE TO GRAYSCALE
# =====

gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

print("Grayscale conversion completed.")
print("Image size:", gray.shape)
print("Himani Chaudhary")
print("CU240250967")
```

Grayscale conversion completed.
Image size: (3584, 5376)
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```
# 3. CONTRAST STRETCHING
# =====

min_pixel = np.min(gray)
max_pixel = np.max(gray)

if max_pixel == min_pixel:
    contrast_stretched = gray.copy()
else:
    contrast_stretched = (
        (gray - min_pixel) /
        (max_pixel - min_pixel) * 255
    ).astype(np.uint8)
    print("Himani Chaudhary")
    print("CU240250967")
```

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```
# 4. HISTOGRAM EQUALIZATION
# =====
```

```
hist_equalized = cv2.equalizeHist(gray)
print("Himani Chaudhary")
print("CU240250967")
```

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```
# 5. CLAHE
# =====

clahe = cv2.createCLAHE(
    clipLimit=2.0,
    tileGridSize=(8, 8)
)

clahe_image = clahe.apply(gray)
print("Himani Chaudhary")
print("CU240250967")
```

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```
# 6. DISPLAY ORIGINAL AND ENHANCED IMAGES
# =====

plt.figure(figsize=(15, 10))

plt.subplot(2, 3, 1)
plt.imshow(gray, cmap="gray")
plt.title("Original Grayscale")
plt.axis("off")

plt.subplot(2, 3, 2)
plt.imshow(contrast_stretched, cmap="gray")
plt.title("Contrast Stretching")
plt.axis("off")

plt.subplot(2, 3, 3)
plt.imshow(hist_equalized, cmap="gray")
plt.title("Histogram Equalization")
plt.axis("off")

plt.subplot(2, 3, 4)
plt.imshow(clahe_image, cmap="gray")
plt.title("CLAHE")
plt.axis("off")

plt.tight_layout()

plt.savefig(
    "contrast_enhancement_results.jpg",
    dpi=300
)

plt.show()
print("Himani Chaudhary")
print("CU240250967")
```

Original Grayscale



Contrast Stretching



Histogram Equalization



CLAHE



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```
# 7. HISTOGRAMS
# =====

plt.figure(figsize=(15, 8))

plt.subplot(2, 2, 1)
plt.hist(gray.ravel(), 256, [0, 256])
plt.title("Original Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.subplot(2, 2, 2)
plt.hist(
    contrast_stretched.ravel(),
    256,
    [0, 256]
)
plt.title("Contrast Stretching Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.subplot(2, 2, 3)
plt.hist(
    hist_equalized.ravel(),
    256,
    [0, 256]
)
plt.title("Histogram Equalization")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.subplot(2, 2, 4)
plt.hist(
    clahe_image.ravel(),
    256,
    [0, 256]
)
plt.title("CLAHE Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

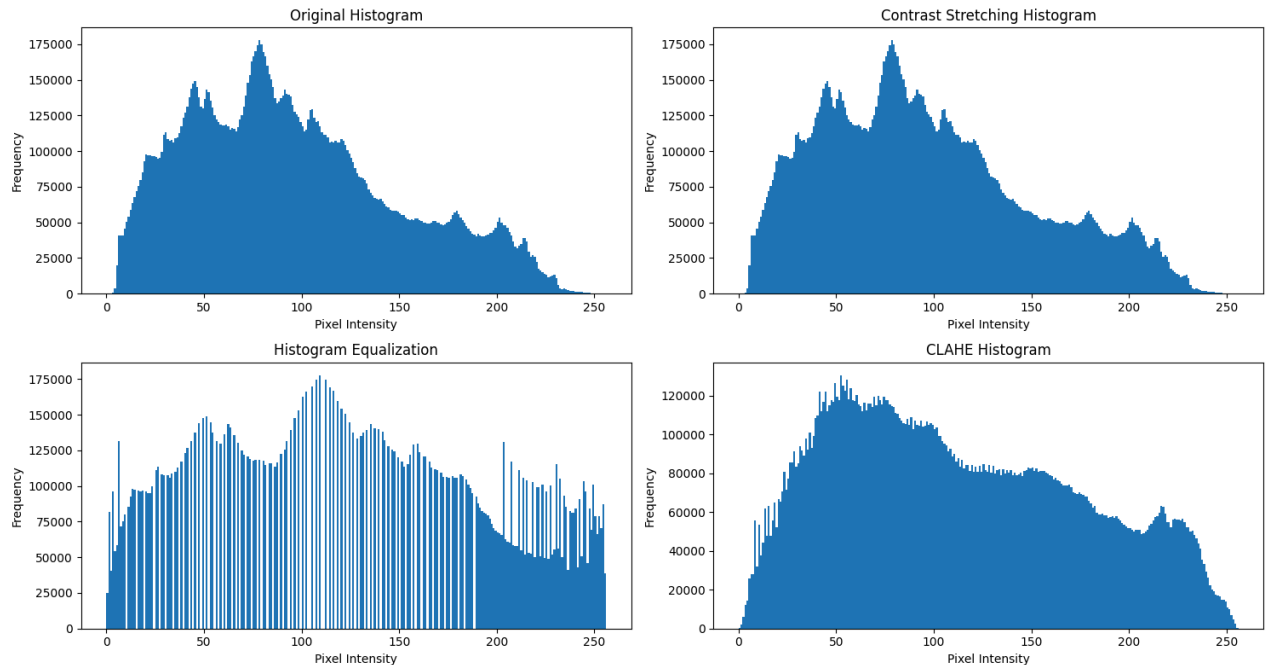
plt.tight_layout()

plt.savefig(
    "histograms.jpg",
    dpi=300
)

plt.show()
```

```
print("Himani Chaudhary")
print("CU240250967")
```

```
/tmp/ipykernel_1388/1730121382.py:7: MatplotlibDeprecationWarning: Passing the range parameter of hist() positionally is deprecated
plt.hist(gray.ravel(), 256, [0, 256])
/tmp/ipykernel_1388/1730121382.py:13: MatplotlibDeprecationWarning: Passing the range parameter of hist() positionally is deprecated
plt.hist(
/tmp/ipykernel_1388/1730121382.py:23: MatplotlibDeprecationWarning: Passing the range parameter of hist() positionally is deprecated
plt.hist(
/tmp/ipykernel_1388/1730121382.py:33: MatplotlibDeprecationWarning: Passing the range parameter of hist() positionally is deprecated
plt.hist(
```



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```
# 8. SAVE ENHANCED IMAGES
# =====

cv2.imwrite(
    "contrast_stretched.jpg",
    contrast_stretched
)

cv2.imwrite(
    "histogram_equalized.jpg",
    hist_equalized
)

cv2.imwrite(
    "clahe_image.jpg",
    clahe_image
)
print("Himani Chaudhary")
print("CU240250967")
```

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```
# 9. FINAL OUTPUT
# =====

print("=====")
print("Processing completed successfully!")
print("=====")

print("Saved files:")
print("1. contrast_stretched.jpg")
print("2. histogram_equalized.jpg")
print("3. clahe_image.jpg")
print("4. contrast_enhancement_results.jpg")
print("5. histograms.jpg")
```

```
print("Himani Chaudhary")  
print("CU240250967")
```

```
=====  
Processing completed successfully!  
=====  
Saved files:  
1. contrast_stretched.jpg  
2. histogram_equalized.jpg  
3. clahe_image.jpg  
4. contrast_enhancement_results.jpg  
5. histograms.jpg  
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```